

Total No. of Questions : 8]

SEAT No. :

PA-1501

[Total No. of Pages : 4

[5926]-121

T.E. (IT)

MACHINE LEARNING

(2019 Pattern) (Semester - I) (314443)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) What do you mean by coefficient of regression? Explain SSE, MSE and MAE in context of regression. [CO2, L3] [5]

b) What is multiple regression? How it is different from simple linear regression [CO2, L1] [5]

c) Consider the following data

The values of x and their corresponding values of y are shown in the table below

- i) Find values of β_0 and β_1 w.r.t. linear regression model which best fits given data.
- ii) Interpret and explain equation of regression line.
- iii) Estimate the value of y for $x = 90$.

	x	y
1	95	85
2	85	95
3	80	70
4	70	65
5	60	70

[CO2, L3]

[7]

OR

P.T.O.

- Q2) a)** Explain under fit, over fit and just fit models for Regression [CO2, L1] [5]
- b) Explain bias-variance dilemma [CO2, L2] [5]
- c) What is univariate and multivariate regression? Explain any three measures of Evaluation of performance of regression model. [CO2, L2] [7]

- Q3) a)** For the given data set apply Naïve Bayes Classifier and predict the Class for weather = Sunny and car = working. [10]

Weather	Car	Class
Sunny	Working	Go-out
Rainy	Broken	Go-out
Sunny	Working	Go-out
Sunny	Working	Go-out
Sunny	Working	Go-out
Rainy	Broken	Stay-home
Rainy	Broken	Stay-home
Sunny	Working	Stay-home
Sunny	Broken	Stay-home
Rainy	Broken	Stay-home

[CO4, L3]

- b) What is decision tree? Explain ID-3 algorithm of Decision tree in detail. [CO4, L2] [8]

OR

- Q4) a)** For the following data calculate weighted average entropy for all features.
 Length = [3,4,5] [2+, 0-] [1+, 3-] [2+, 2-]
 Gills = [Yes, No] [0+, 4-] [5+, 1-]
 Beak = [Yes, No] [5+, 3-] [0+, 2-]
 Teeth = [many, few] [3+, 4-] [2+, 1-]
 [CO4, L3]

[10]

- b) Define and Explain following terms [8]
- Bayesian Network
 - Advantages and disadvantages of Naïve Bayes Classifier [CO4, L2]

- Q5) a)** Find all association rules using apriori algorithm in the following database in the following database with minimum support = 2 and minimum confidence = 65%. **[10]**

Transactions	Data Items
T1	Pen, Pencil, Notebook
T2	Pencil, File
T3	Pen, Pencil, Notebook, File
T4	Pen, Notebook
T5	Pencil, Scale, File
T6	Pencil, Scale
T7	Pen, Pencil, Scale

[CO5, L3]

- b) What is use of K-means algorithm? Explain Centroid and medoid? Explain different types of distances measures. **[CO5, L2]** **[8]**

OR

- Q6) a)** Explain following Terms **[8]**

- Rule
- Support
- Lift
- Confidence

[CO5, L2]

- b) Apply KNN on the following data and classify the new sample (3,7) to the respective class. **[10]**

X	Y	Class
7	7	Pass
7	4	Pass
3	4	Fail
1	4	Fail
4	3	Fail
6	7	Pass
3	7	?

What will be the effect on output if $k = 3$ and $k = 5$?

[CO5, L3]

- Q7) a)** With the help of suitable diagram explain Biological Neuron. **[CO6, L3]** **[6]**
- b)** What is the use of activation function in Neural Network? Explain any two activation functions in detail **[CO6, L2]** **[6]**
- c)** What is deep learning? Explain different applications of deep learning. **[CO6, L1]** **[5]**

OR

- Q8) a)** What is perceptron? Explain multilayer perceptron in detail. **[CO6, L3]** **[6]**
- b)** Write a note on following activation functions. **[6]**
- i) Sigmoid
 - ii) Tanh
 - iii) ReLU **[CO6, L2]**
- c)** What is ANN? Explain McCulloch Pitts Neuron **[CO6, L2]** **[5]**



Total No. of Questions: 8]

SEAT No. :

P403

[6003]-703

[Total No. of Pages : 4

T.E. (I.T.)

MACHINE LEARNING

(2019 Pattern) (Semester-I) (314443)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1) a) State and explain need of Regression analysis. [CO2, L2] [4]
b) How gradient descent does helps to optimize linear regression model? [CO2, L2] [6]
c) What are the different ways to prevent overfitting [CO2, L2] [8]

OR

- Q2) a) What are different cost functions to access the performance of linear Regression model? In the given Dataset the outliers represent anomalies. Which cost function will be more suitable and why? [CO2, L3] [5]
b) Define of Multivariate Regression and State advantages and disadvantages of Multivariate Regression. [CO2, L2] [5]

Consider the following data

	Prize in Rs	Amount Demanded
1	10	40
2	11	38
3	16	48
4	18	40
5	20	60

c)

- i) Find values β_0 and β_1 w.r.t. linear regression model which best fits given data.
 - ii) Interpret and explain equation of regression line.
 - iii) Estimate the likely demand when the price is Rs. 15.
- [CO2, L3] [8]

P.T.O.

- Q3) a)** Consider following data. Which feature will be selected as a root node? Use Information Gain. Played football is dependent feature. [CO4, L3]

[10]

Outlook	Temperature	Humidity	Wind	Played football(yes/no)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

- b)** Define and Explain following terms **[7]**

- Bayesian Network
- Advantages and disadvantages of Naïve Bayes Classifier [CO4, L2]

OR

- Q4) a)** For the given data set apply Naive Bayes Classifier and predict the Class for Type of Type of family structure = Single Parent, Age group = Young, and Income status=Low [CO4, L3] **[10]**

Type of family structure	Age group	Income status	will they buy a car?
Nuclear	Young	Low	Yes
Extended	old	Low	No
Childless	Middle-aged	Low	No
Childless	Young	Medium	Yes
Single Parent	Middle-aged	Medium	Yes
Childless	Young	Low	No
Nuclear	Old	High	Yes
Nuclear	Middle-aged	Medium	Yes
Extended	Middle-aged	High	Yes
Single Parent	Old	Low	No

b) Define and explain following terms [7]

- i) Minority Class
- ii) Gini Index
- iii) Entropy
- iv) Information Gain [CO4, L2]

Q5) a) Find all association rules in the following database in the following database with minimum support = 2 and minimum confidence = 75% [CO5, L3] [10]

Transactions	Data Items
1	Bread, Milk, Diaper
2	Bread, Milk, Diaper, Coke
3	Diaper, Beer, Eggs
4	Bread, Milk, Coke

b) State & explain with appropriate example different types of linkage use in clustering. [CO5, L2] [8]

OR

Q6) a) Explain following Terms [8]

- i) Rule
 - ii) Support
 - iii) Lift
 - iv) Confidence
- [CO5, L2]

- b) Apply KNN on the Following data. Find class of person whose height is 170 cm and weight is 57 kg. Consider value $K = 5$ and use Euclidian distance formula. [CO5, L3] [10]

Height (CM)	Weight (KG)	Class
167	51	Underweight
182	62	Normal
176	69	Normal
173	64	Normal
172	65	Normal
174	56	Underweight
169	58	Normal
173	57	Normal
170	55	Normal

- Q7)** a) With the help of suitable diagram explain Biological Neuron. [CO6, L3] [5]
 b) Explain the architecture of feed forward neural network. State its limitations. [CO6, 12] [7]
 c) What is deep learning? Explain different applications of deep learning [CO6, L1] [5]

OR

- Q8)** a) What is perceptron? Explain multilayer perceptron in detail. [CO6, L3] [5]
 b) Explain why we use non-linearity function? State and explain three types of neurons that add non-linearity in their computations [CO6, 12] [7]
 c) What is ANN? Explain McCulloch Pitts Neuron [CO6, L2] [5]



Total No. of Questions : 8]

SEAT No. :

PB-3860

[Total No. of Pages : 4

[6262]123

T.E. (IT)

MACHINE LEARNING

(2019 Pattern) (Semester - I) (314443)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) State and explain need of Regression analysis. [4]
b) How gradient descent does helps to optimize linear regression model? [6]
c) What are the different ways to prevent overfitting and underfitting. [8]

OR

- Q2)** a) What are different cost functions to access the performance of linear Regression model? In the given Dataset the outliers represent anomalies. Which cost function will be more suitable and why? [5]
b) Define of Multivariate Regression and State advantages and disadvantages of Multivariate Regression. [5]
c) Consider the following data : [8]

	Prize in Rs	Amount Demanded
1	10	40
2	11	38
3	16	48
4	18	40
5	20	60

- i) Find values of β_0 and β_1 w.r.t. linear regression model which best fits given data.
- ii) Interpret and explain equation of regression line.
- iii) Estimate the likely demand when the price is Rs.15.

P.T.O.

- Q3) a)** Consider following data. Which feature will be selected as a root node? Use Information Gain. Played football is dependent feature. **[10]**

Outlook	Temperature	Humidity	Wind	Played football (Yes/No)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

- b)** Define and Explain following terms **[7]**

- Bayesian Network
- Advantages and disadvantages of Naïve Bayes Classifier

OR

- Q4) a)** For the given data set apply Naive Bayes Classifier and predict the Class for Type of family structure=Single Parent, Age group=Young, and Income status=Low **[10]**

Type of family structure	Age group	Income status	Will they buy a car
Nuclear	Young	Low	Yes
Extended	Old	Low	No
Childless	Middle-aged	Low	No
Childless	Young	Medium	Yes
Single Parent	Middle-aged	Medium	Yes
Childless	Young	Low	No
Nuclear	Old	High	Yes
Nuclear	Middle-aged	Medium	Yes
Extended	Middle-aged	High	Yes
Single Parent	Old	Low	No

- b) Define and explain following terms [7]
- Minority Class
 - Gini Index
 - Entropy
 - Information Gain

- Q5) a)** Find all association rules in the following database in the following database with minimum support = 2 and minimum confidence = 75%. [10]

Transactions	Data Items
1	Bread, Milk, Diaper
2	Bread, Milk, Diaper, Coke
3	Diaper, Beer, Eggs
4	Bread, Milk, Coke

- b) State & explain with appropriate example different types of linkage use in clustering. [8]

OR

- Q6) a)** Explain following Terms [8]

- Rule
- Support
- Lift
- Confidence

- b) Apply KNN on the Following data. Find class of person whose height is 170 cm and weight is 57 kg. Consider value $K = 5$ and use Euclidian distance formula. [10]

Height (CM)	Weight (KG)	Class
167	51	Underweight
182	62	Normal
176	69	Normal
173	64	Normal
172	65	Normal
174	56	Underweight
169	58	Normal
173	57	Normal
170	55	Normal

- Q7)** a) With the help of suitable diagram explain Biological Neuron. [5]
b) Explain the architecture of feed forward neural network. State its limitations. [7]
c) What is deep learning? Explain different applications of deep learning. [5]

OR

- Q8)** a) What is perceptron? Explain multilayer perceptron in detail. [5]
b) Explain why we use non-linearity function? State and explain three types of neurons that add non-linearity in their computations. [7]
c) What is ANN? Explain McCulloch Pitts Neuron. [5]

Total No. of Questions : 8]

SEAT No. :

P-7619

[Total No. of Pages : 3

[6180]-139

T.E. (Information Technology)

MACHINE LEARNING

(2019 Pattern) (Semester - I) (314443)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1) a) Compare univariate and multivariate linear regression. [CO3,L2] [6]
- b) Describe the tradeoff between bias and variance using dart example. [CO3,L1] [5]
- c) Compute the R-Square error of the best fit line " $Y = -0.94X + 43.7$ " for below input instance space [CO3, L3] [7]

Sr.No.	X	Y
1	3	40
2	10	35
3	11	30
4	15	32
5	22	19
6	22	26
7	23	24
8	28	22
9	28	18
10	35	6

OR

P.T.O.

- Q2)** a) Explain gradient descent technique for optimization in linear regression with example. **[CO3, L1]** [6]
 b) Explain the cost function used to evaluate the performance of regression. **[CO3, L3]** [6]
 c) What is least square method? Explain least square method in the context of regression. **[CO3, L1]** [6]

- Q3)** a) Describe Bayesian network in short for learning and inferences. **[CO4, L3]** [8]
 b) Explain naïve bayes algorithm? For below dataset compute the posterior probability of banana class using Bayes rule. **[CO4, L3]** [7]

Fruit	Yellow	Sweet	Long	Total
Orange	350	450	0	650
Banana	400	300	350	400
Other	50	100	50	150
Total	800	850	400	1200

- c) Write any four applications of naïve Bayes classifier. **[CO4, L1]** [2]

OR

- Q4)** a) Explain ID-3 decision tree algorithm in detail with example. **[CO4, L2]** [8]
 b) Explain the following measures of impurity with example. **[CO4, L2]** [9]
 i) Information Gain ii) Gini Index iii) Entropy

- Q5)** a) Explain K-Nearest Neighbor algorithm with example. **[CO5, L2]** [7]
 b) Suppose we have the following dataset that has various transactions and from this dataset, we need to find the frequent item sets and generate the association rules using the Apriori algorithm. Given minimum support = 2 and minimum confidence = 50%. **[CO5, L4]** [8]

TID	ITEM SET	TID	ITEM SET
T1	A,B	T6	B,C
T2	B,D	T7	A,C
T3	B,C	T8	A, B, C, E
T4	A, B, D	T9	A, B, C
T5	A, C		

c) Explain any one of the following terms: [CO5, L2] [3]

i) Medoid OR ii) Dendrogram

OR

Q6) a) Cluster the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9). Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2). The distance function between two points $a = (x_1, y_1)$ and $b = (x_2, y_2)$ is defined as $P(a, b) = |x_2 - x_1| + |y_2 - y_1|$. Use K-Means Algorithm to find the three cluster centers after the first iteration. [CO5, L3] [9]

b) Explain the Apriori algorithm in brief. Explain the following performance measure of association rule mining. [CO5, L2] [6]

1) Support 2) Confidence 3) Lift

Explain any one of the following distance metrics with example. [CO5, L1] [3]

1) Euclidean Distance 2) Manhattan Distance

3) Hamming Distance

Q7) a) Explain perceptron learning algorithm? Describe shortly about how learning parameters are updated in multi-layer perceptron. [CO6, L3] [8]

b) Explain Sigmoid, Tanh and Relu activation functions in detail. [CO6, L2] [9]

OR

Q8) a) Explain the simulation of AND gate using McCulloch Pitts Neuron? What is vanishing gradient descent problem. [CO6, L2] [9]

b) Explain what is Deep Learning and its different architectures? State the [CO6, L2] various applications of deep learning. [8]



Total No. of Questions : 8]

PC1803

SEAT No. :

[Total No. of Pages : 4

[6353]-122

T.E. (IT)

MACHINE LEARNING

(2019 Pattern) (Semester - I) (314443)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4 Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) What is linear Regression? When is it suitable to use linear regression over classification? [6]

b) What is bias-variance? Give its impact in choosing the best model. [5]

c) Explain evaluation metrics or loss functions for regression. [6]

OR

Q2) a) Explain gradient descent with respect to linear regression. [6]

b) Explain under fit, over fit and just fit models for Regression. [5]

c) What is regression? Explain univariate regression and multivariate regression. [6]

Q3) a) What is decision tree? Explain ID-3 algorithm of Decision tree in detail. [8]

b) Define [4]

i) Conditional Probability

ii) Posterior Probability

c) Explain in brief the Bayesian network for learning and inference. [6]

OR

P.T.O.

- Q4) a)** For given dataset predict the class of new test instance using naïve Bayes classifier. Given a new instance as:

(Outlook = Sunny; Temperature = Cool; Humidity = High; Wind = Strong)

What will be the value of PlayTennis? **[12]**

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

- b) Explain the measures of impurity (Information Gain, Gini Index, and Entropy). **[6]**

- Q5) a)** Using k-means algorithm cluster the following eight data points (with (x, y) representing locations) into three clusters ($C1 = (2,5)$, $C2 = (5,8)$, $C3 = (1,2)$) [12]

X:	2	2	8	5	7	6	1	4
Y:	10	5	4	8	5	4	2	9

- b) Explain the following terms: [6]

- i) Support
- ii) Confidence
- iii) Lift

OR

- Q6) a)** Consider the following sample dataset of students. Using K-Nearest-Neighbour classifier predicts the result of new student S_{new} (new test instance). (use $k=3$ and Euclidean Distance measure) [10]

Student	Rating by Internal Examiner (X_{i1})	Rating by External Examiner (X_{i2})	Result (Y)
S1	7	7	PASS
S2	7	4	PASS
S3	3	4	FAIL
S4	1	4	FAIL
S_{new}	3	7	??

- b) Explain Hierarchical clustering with suitable example. [8]

- Q7)** a) With the help of suitable diagram explain architecture of Artificial Neural Network. [6]
- b) What is the use of activation function in Neural Network? Explain any two activation functions in detail. [6]
- c) Explain deep learning with applications. [5]

OR

- Q8)** a) Explain Perceptron training algorithm used for linear classification. [6]
- b) Write a note on following activation functions. [6]
- i) Sigmoid
 - ii) Tanh
 - iii) ReLU
- c) How artificial neuron is inspired by biological neuron. [5]

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Total No. of Questions : 8]

SEAT No. :

PD-4324

[Total No. of Pages : 2

[6403]-122
T.E. (I.T)
MACHINE LEARNING
(2019 Pattern) (Semester - V) (314443)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Define Regression and explain types of regression with proper examples. **[6]**
b) Explain Generalization Issues with Overfitting, Underfitting and Bias, Variance. **[5]**
c) Explain Least-Square Method for finding values of the regression coefficients **[6]**

OR

- Q2)** a) Explain Simple Linear Regression with Gradient Descent Algorithm. **[6]**
b) Explain Multivariate Regression with model representation. **[5]**
c) Write short note on Vapnik-Chervonenkis dimension. **[6]**

- Q3)** a) What is Decision Tree? Explain Concepts and Terminologies. **[8]**
b) Explain Bayes Rule & Naive Bayes Classifier. **[4]**
c) Define Gini Index, Information gain and Entropy. **[6]**

OR

- Q4)** a) Explain feature tree & comment on best split algorithm. **[12]**
b) Explain Conditional Probability, Joint Probability and Probability Density Function. **[6]**

P.T.O.

- Q5) a)** Apply Apriori algorithm for following set of transactions and find all the association rules with min support = 1 and min confidence = 60%. **[12]**

Transaction ID	Transactions
1	1,3,4
2	2,3,5
3	1,2,3,5
4	2,5

- b) Define and explain various Distance Metrics. **[6]**

OR

- Q6) a)** Suppose the task is to cluster points into three clusters using K-means clustering algorithm, where the points are A 1(2, 10), A2(2, 5), A3(8, 4), B 1(5, 8), B2(7, 5), B3(6, 4), C 1(1, 2), C2(4, 9). Find the members of these 3 clusters after 2 iterations. **[10]**

- b) What is distance metric? Explain any three distance measures. **[8]**

- Q7) a)** Explain different Activation Functions and it's types. **[6]**

- b) Explain Multi-layer Perceptron with example. **[6]**

- c) Write a note on Loss function- Mean Square Error. **[5]**

OR

- Q8) a)** Explain Perceptron with it's Learning Algorithm. **[6]**

- b) Write a note on ANN. **[6]**

- c) Elaborate Deep Learning in detail. **[5]**

